



Interactive Method to Elicit Local Causal Knowledge for Creating a Huge Causal Network

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Abstract. Effective disaster drills and exercises require appropriate scenarios reflecting concrete disaster situations. It is however not easy to manually create such a scenario with enough details and validity, because it is fundamental difficult to comprehensively predict and assume disaster situations that may occur in various phases through a chain of causality from the primary damage. In order to make a scenario creation easier and more efficient, some support tools are necessary, in particular for predicting what kind of situations will happen through a causal chain from a base disaster assumption. In this paper, we proposed a simple and practical causal model consisting of three elements: cause, precondition, and effect, which can capture indirect causal relationships between two events by introducing the concept of preconditions. We also developed an interactive method with a GUI to elicit causal knowledge about disaster situations based on the model. Users can enter possible events that can occur in a disaster as well as countermeasures against those events by answering the questions presented on the GUI. Then the entered sentences are processed to identify causal elements automatically by a newly developed NLP techniques, and finally those elements are integrated into the database. The proposed method still has a room for improvement, however its performance is satisfying and can be expected to be utilized as a technical base for the creation of effective disaster scenarios.

Keywords: Causal model · Knowledge elicitation · Scenario creation

1 Introduction

In recent years, the concept of resilience has attracted attention in the field of disaster prevention. Bruneau et al. defined resilience as a generic term for ability to reduce failure probabilities, ability to reduce consequences from failures, and ability to reduce time to recovery [1]. In order for that, it is necessary for organizations to quickly establish an initial response system in a disaster and make appropriate and prompt decisions based on uncertain information. For developing this capacity to respond to disasters, it is not enough to simply read and understand a disaster response manual, but disaster drills and exercises are necessary. Effective drills and exercises require appropriate disaster scenarios that prescribe concrete disaster situations including

background situations of the organization, damage assumptions, the constraints caused by the damage, and so on.

It is however not easy to manually create such a disaster scenario with enough details and validity, in particular for people without special knowledge and experience. This is not only due to insufficient time or lack of knowledge on disasters, but also due to the fundamental difficulty for individuals to comprehensively predict and assume the extent of damage effects that may occur in various phases through a chain of causality. In disasters, different damages and events occur in a chain reaction from a certain damage and event, resulting in a chain failure caused by the dependency of the social systems and infrastructure, and unexpected events such as a chain reaction of events within a limited spatial scope.

Limousin et al. developed a support tool to create disaster scenarios using databases and matrices [2]. In this tool, cross impact analysis methodology is used to determine how relationships between events may impact resulting events. Since it is difficult for non-experts to use cross impact methodology, it is not easy to create scenarios considering the extent of damage effects that may occur in various phases through a chain of causality by using this tool. Judek et al. improved a crisis simulation approach called iCrisis and enabled the preparation of disaster scenarios considering cascading effects [3]. Using this tool, scenarios considering causal relationships can be created. However, this tool consists only of checklists and guidance, making it is still not easy for non-experts to create appropriate disaster scenarios.

To predict and describe a chain of causality is one of the biggest obstacles for non-experts in creating effective disaster scenarios. Therefore, it is necessary to support this by for example eliciting and reusing the knowledge obtained from many people.

There are two basic approaches for eliciting knowledge. One approach is automatic knowledge elicitation using AI techniques with natural language processing (NLP). There are many researches on knowledge elicitation from Web by text mining. This approach however has several problems. One is that current natural language processing techniques does not have enough accuracy. Another problem is that the information on the Web is more general and less specific to individual organizations although local and specific knowledge is required for individual organizations. Another approach is using manual methods such as interview, questionnaire, and model-based analysis. However, these methods require much time and effort, in particular in analyzing data and converting the knowledge into a reusable form. In addition, there are no good theoretical model for describing causal knowledge. For example, a simple cause and effect model is too simple to describe causal relationships behind an event. On the other hand, if we try to describe causal relationships in more detail, such as FRAM, a functional resonance analysis method, the model becomes complicated and difficult for non-expert to understand and use.

From this background, in this paper we propose a causal model consisting of three elements, cause, condition and effect, which is simple and practical to describe causal relationships. To solve the problems of the current manual and automatic knowledge elicitation method, we propose and develop an interactive method to elicit causal knowledge.

2 Causal Model

We propose a new causal model consisting of cause, preconditions, and effects.

FRAM describes six aspects of causal relationships, which makes the method difficult for non-experts to understand and use it. On the other hand, there are many phenomena that cannot be explained and understood by a simple cause and effect model. One of the example of such phenomena that cannot be explained by a simple model is that a medical equipment cannot be used due to blackout in a building with inadequate emergency power supply. “Blackout” is a direct factor that makes an effect (in this case, “medical equipment cannot be used”), but “inadequate emergency power supply” is not a direct factor for the effect. However, when “blackout” occurs, and if there is “inadequate emergency power supply”, the effect occurs. But if there is not “inadequate emergency power supply”, the effect does not occur. In this study, a direct causal factor is defined as “cause”, while an event that is not a direct factor but is necessary for an effect to occur is defined as “precondition”. In the above example, the cause is “blackout” and the precondition is “inadequate emergency power supply”.

In our model, a precondition can cover four of the six aspects of function of FRAM (Time, Control, Precondition, and Resource). The remaining two aspects (Input and Output) correspond to a cause and an effect. This causal model consisting of causes, conditions, and effects is a simpler and more practical than FRAM. At the same time, this model has more descriptive power than a simple causal model consisting only of causes and effects. It is still difficult to consider and pre-identify preconditions behind events occurred in a disaster, which is the target causal relationships in this study. Therefore, we focus on countermeasures instead of preconditions, because countermeasures can be relatively easily elicited as ideas for preventing potential harmful events from occurring when a disaster occurs. In other words, we can say that countermeasures are interpretations of preconditions from different viewpoints. In the above example, the precondition is “inadequate emergency power supply” and the countermeasure is “installing an adequate emergency power supply”.

3 Method to Elicit Causal Knowledge

We are developing an interactive method to elicit causal knowledge which describe the causal relationships behind events under disaster situations and to utilize the knowledge for creating detailed disaster scenarios. The knowledge elicited by the method is converted and stored in a database based on the causal model described in the previous section. This interactive method provides a GUI to ask questions on disaster situations and enter the answers, which is designed and developed for Japanese language. This method is intended to be used in a workshop.

At the beginning of a workshop, participants are asked to discuss possible events as well as their effects that can occur in a disaster and input them using the GUI, which aims to elicit causal knowledge. In order to enhance the usability and to avoid narrowing the scope of thinking, the participants can input any possible events that can occur in a disaster in free formants. Causal chains are extracted from input sentences automatically by the following two methods. As shown in Fig. 1, several causal

candidates estimated by NLP are displayed on the GUI. First participants are asked to select whether input sentence has a causal relationship or not, then they are asked to select the appropriate cause and effect. If there is no appropriate candidate presented on the GUI, then participants are asked to enter an appropriate causal relationship manually. These procedures should be repeated until every conceivable event in a disaster is entered. The GUI then displays the previously entered pairs of causes and effects one by one. Participants are also asked to enter as many countermeasures as possible to address these causes and effects. As described above, knowledge on countermeasures is elicited instead of preconditions in our method. This is not only because it is easier to come up with countermeasures than preconditions, but also because considering countermeasures becomes a good opportunity for learning about disaster response.

Please select the appropriate one.

☒ With causal relationship

Cause

☒ 窓が割れる ☐ 強いドアにする ☐ 開けられなくなる ☐ 窓が割れたため ☐ 窓が割れた

Effect

☐ 先に開けておく ☒ けがをする ☐ けがをした ☐ ためけがをした

☐ Without causal relationship

☒ けがをする ☐ 窓が割れる ☐ 強いドアにする ☐ 窓が割れたためけがをした

☐ There is nothing appropriate.

Next

Fig. 1. GUI screen

3.1 Method to Extract Causal Relationship

This section describes the algorithms used in the method, especially for natural language processing. We incorporated a method developed by Sakaji et al. (hereinafter referred to as Method A) [5, 6] with a newly developed method (hereinafter referred to as Method B) to enhance the ability for knowledge elicitation. The purpose of Method A is to extract only certain causal relationship in the text, so it is not possible to comprehensively extract possible causalities. This is because causal relationship is often expressed in Japanese without using causal words such as conjunction. Therefore, Method B is developed to extract causal relationship that cannot be extracted by Method A.

Method A uses the sentence structure and clue expressions to determine and extract causal relationship. A clue expression, such as “*No-de*” in Japanese or “as” in English, is an expression that is an important clue to extract the cause and effect. Method A first looks for a clue expression in a target sentence. When a clue expression is found in the sentence, then a dependency analysis is conducted to determine the sentence structure. Dependency analysis is a type of syntax analysis that divides a sentence into morphemes and analyzes the modification relationships between morphemes. In this study, CaboCha [7] was used for dependency analysis. The sentence is categorized into five patterns based on the sentence structure. These five patterns are shown in Fig. 2. Based on them, the causal relationship is identified and extracted.

In Method B, a causal relationship is estimated and extracted by dividing an input sentence into several simple sentences. Although the original definition of a simple sentence means a sentence with only one set of a subject and a predicate, a sentence with a noun clause such as “I hear the news that an earthquake has occurred” is also treated as a simple sentence in Method B. In Japanese, noun clauses often play an auxiliary role in sentences rather than representing a single event, so Method B treats noun clauses as single nouns. In many Japanese sentences that express causal relationship without using clue expressions, the event described earlier tends to be a “cause” and the event described later tends to be an “effect”. Therefore, in Method B, a preceding simple sentence in an input sentence is identified as a “cause” and the following simple sentence as an “effect”. If an input sentence is divided into three sentences, it is assumed that the first sentence is a cause of the middle sentence and the middle sentence is a cause of the last sentence. It is the same even when it is divided into four or more simple sentences. The details of the algorithm is shown in Fig. 3. First, dependency analysis is performed using CaboCha. Next, a predicate is searched, and a modifier of the predicate is confirmed. If the modifier is not a noun, that is, it is not a noun clause, the input sentence is divided. If it is a noun, the input sentence is not divided. This procedure is done for all predicates, and causal relationship is extracted from an input sentence. While Method B can extract more causalities than Method A, it is more likely to extract wrong causalities that do not actually contain causal relationship. Therefore, in this study, as described above, causal candidates extracted by Method A and Method B are presented to participants, and they select the correct causal relationship in order to solve this problem.

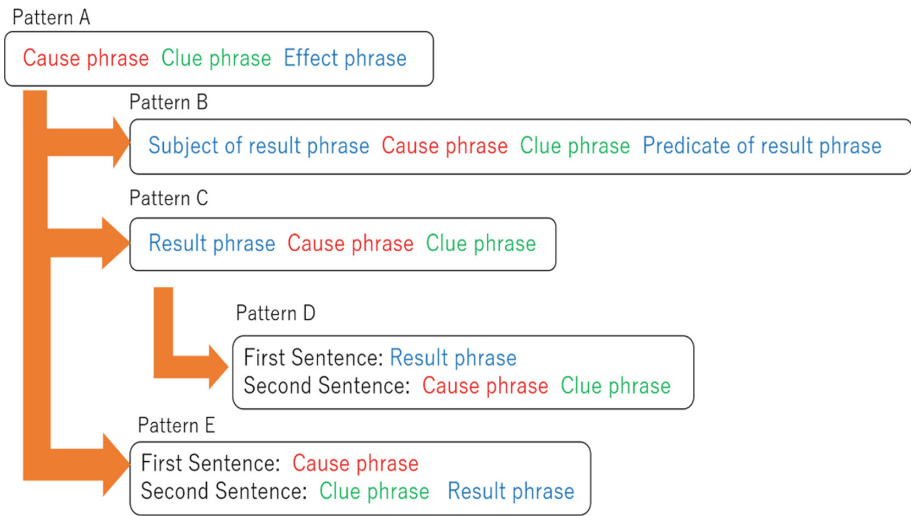


Fig. 2. An association chart of patterns [5]

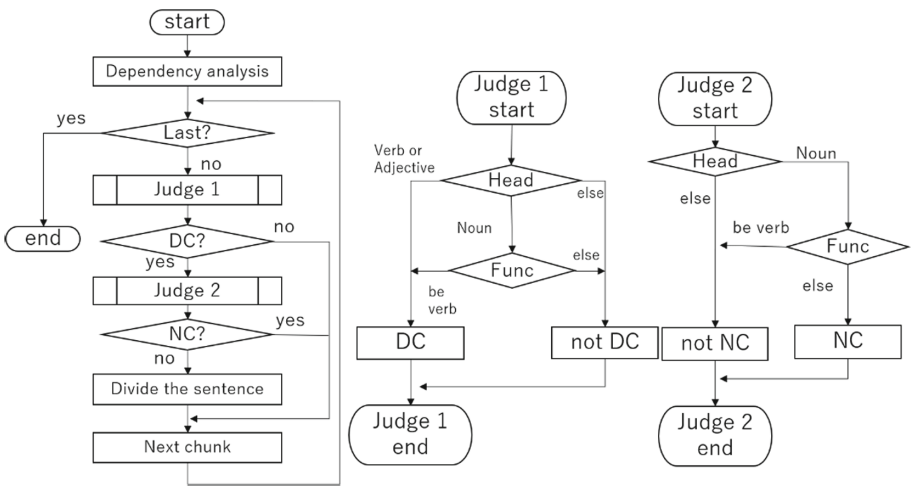


Fig. 3. Flowchart of NLP

3.2 Prevention of Duplication

Different people use different expressions for the same phenomenon. Some people describe “blackout” as “blackout occurs” and others describe “electricity stops”. It is inconvenient if events with different representations are stored as different data in the database. In this study, therefore it is judged whether they are same event or not by evaluating the similarity of sentence vectors obtained by Word2Vec which is a model for converting words into discrete vectors. Since verb, adjective, and noun play main

role in the meaning of a sentence in Japanese language, only vectors of verb, adjective, and noun are used to calculate sentence vectors. If distance between the two sentence vectors is closer than a fixed value, the sentence is determined to have the same meaning. The detailed procedure is as follows.

1. The target sentence (s) is morphemically analyzed.
2. Using a learned model, the vectors of morphemes are calculated.
3. A weighted average of these vectors is calculated to be the vector of s .
4. A vector data set (D_v) is created by repeating step 1 to 3 for a sentence group (D) collected in the past.
5. Steps 1 to 3 are performed on the new sentence (S), and a vector (V) is calculated.
6. Distance between V and the vectors in D_v is calculated, and sentences whose distance is less than a certain value is selected and displayed on the GUI.
7. Participants of the workshop select appropriate one. If there is no appropriate one, S is added to D .

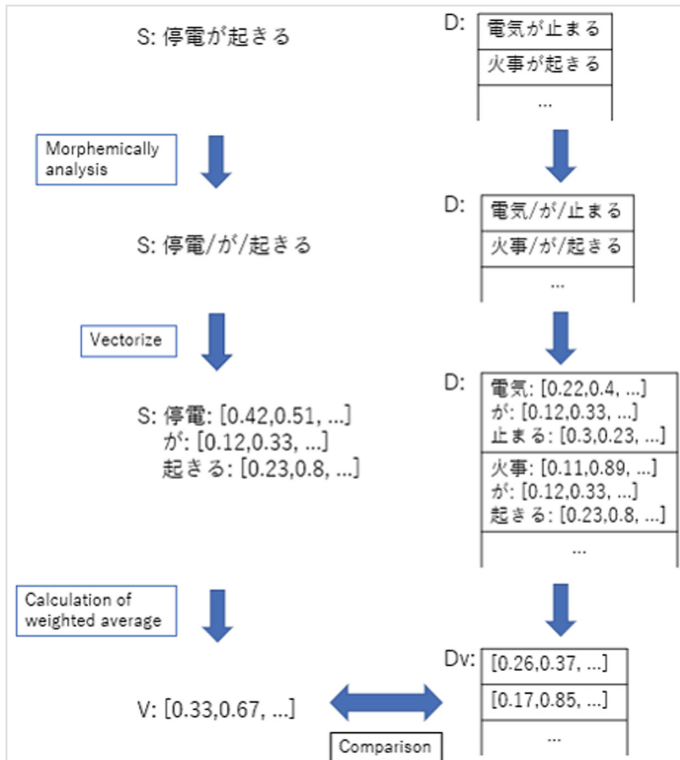


Fig. 4. Flow of similarity evaluation (Color figure online)

3.3 Verification

An experiment was conducted to verify the proposed method. “Kyoto University Web Documentation Lead Corpus” [8] was used as the test data. This corpus is a text corpus in which various language information, such as semantic relations, is added manually to the lead three sentences of various Web documents. In the verification, 100 sentences randomly selected from 628 sentences which contain causal relationships. Method A and Method B were applied to the selected sentences, and the correctness was judged manually.

Table 1 shows results of the experiment. Method A identified correct causal relationships in 46 sentences, while Method B did 63 sentences. Both methods failed to identify correct causal relationships in the same 14 sentences.

Table 1. Results of the verification experiment

Method A	Method B	Combining Method A and B
46	63	87

4 Preliminary Experiment

A preliminary experiment using the propose method was carried out to elicit causal knowledge from two participants. In the experiment, the knowledge about events which can occur in a room in a big earthquake was tried to be elicited. The procedure was as described in Sect. 3. The causal knowledge obtained from the two participants are shown in Fig. 4. The red node in the center represents the occurrence of disaster, in this time an earthquake, blue nodes represent events and damages that can be caused by the disaster, and small purple nodes represent preconditions for the event connected by an edge. An edge represents a causal relationship, and a tail of an edge between events represents a “cause” and a head of an edge represents an “effect”. Where an event or disaster leads to an event via a precondition, an upstream event or disaster represent a “cause” and a downstream event represents an “effect”.

Twenty events and 15 preconditions (countermeasures) were elicited in the experiment. The knowledge obtained from the first participant (hereinafter referred to as person A) and that from the second participant (hereinafter referred to as person B) were integrated and stored in a database. Person A thought that the earthquake would cause a fire, but person B thought that the gas stove would fall and the carpet would catch fire, then a fire would occur. In this way, by eliciting knowledge from multiple people, one person’s knowledge can be supplemented by another person’s knowledge. In this experiment, local knowledge of the room, such as a gas stove near a carpet, was elicited. If this method is applied to a specific facility or organization, it is expected to elicit local causal knowledge around that facility or organization (Fig. 5).

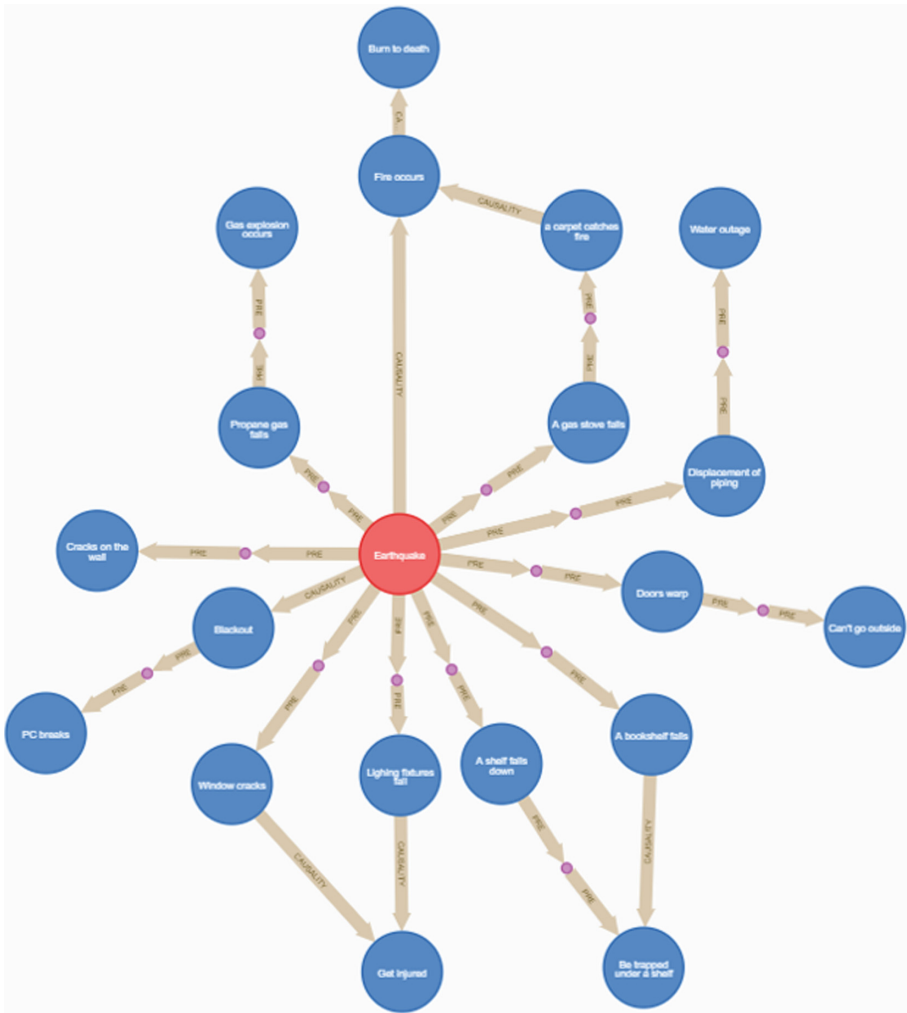


Fig. 5. Visualization of the causal knowledge

On the other hand, this method still has some limitations. One limitation is that the method does not have a function to help users to expand the imagination of events that can occur in a chain via three or more events ahead triggered by some other events. For example, person A thought that water outage would occur due to the displacement of piping, but neither A nor B thought about possible events that would occur due to water outage. In order to elicit further or deeper causal knowledge, the method need to provide some support, for example presenting a causal network to the users while they are working in a workshop. By presenting the causal network and let the users focus on the node where a water outage occurs, they are expected to be evoked to start thinking about what will happen when a water outage occurs.

5 Conclusion

We proposed a simple and practical causal model consisting of three elements: cause, precondition, and effect. This model can capture indirect causal relationships between two events by introducing the concept of preconditions. We also developed an interactive method with a GUI to elicit causal knowledge about disaster situations based on the model, aiming at developing a support tool for non-experts to easily create an effective disaster scenario considering causal chains behind disaster events. Users can enter possible disaster events that can occur in a disaster as well as countermeasures against those events by answering the questions presented on the GUI. Then the entered sentences are processed to identify causal elements automatically by NLP techniques, and finally those elements are integrated into the database. Through this process, the user can modify the causal elements and their relations that were not correctly identified, which can enhance the accuracy and integrity of the causal data.

We first conducted a performance test of the newly developed NLP method to elicit causal relationship and confirmed that the proposed method outperformed the former method.

We also conducted a preliminary experiment to verify and validate the performance of the proposed method and confirmed that the method could correctly identify causal elements and successfully integrated them into the database. The proposed method still has a room for improvement, however its performance is satisfying and can be expected to be utilized as a technical base for the creation of effective disaster scenarios.

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